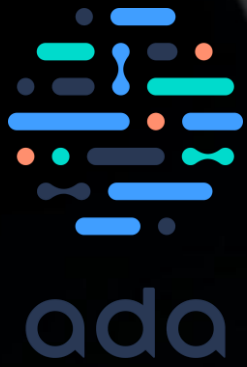


Analyzing Real-World Usage of AI: The Ada Health Chatbot Case Study

Human-Centered AI in Modern Medicine

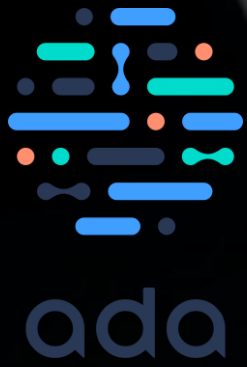


CASE STUDY OVERVIEW

What is Ada Health? A popular, physician-led AI chatbot for symptom checking.

The Mission: Providing personalized health assessments to millions globally.

The Role: Acts as a "digital front door" to the healthcare system. (Ada Health, n.d.)



SYSTEM CREATION

Human-Centered Design (HCD): Built using iterative loops focused on patient empathy.

Learning Sources: Real-world patient/doctor interviews and historical medical data.

Co-Creation: Ongoing collaboration between developers, clinicians, and patients to ensure clinical safety (**Ada Health, n.d.**)



ETHICAL DILEMMAS

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Professional Friction: Physicians worry about AI replacing the "gut feeling" developed through years of training.

Algorithmic Bias: Risk of misdiagnosis if training data is narrow (e.g., assuming lung cancer is only for smokers).

Data Privacy: The ethical burden of protecting sensitive medical information.



USER NEEDS & SOCIETAL IMPACT

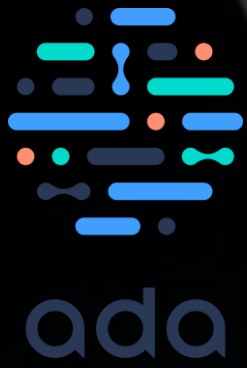
User Needs are guided by HCD principles of empathy, trust, and utility (Design Council, 2019).

User Needs are:

- Functional: Accurate symptom triage.
- Emotional: Anxiety reduction and reassurance.
- AI-Specific: Transparency in how conclusions are reached.

Societal Impact

- Pro:** Improves health literacy by simplifying "medicalese."
- Con:** "Substitution risk"—users might skip real doctors if they trust the AI too much (Ada Health, n.d.).



NATURAL LANGUAGE PROCESSING (NLP)

NLP converts "my head is thumping" into structured clinical data (**Ada Health, n.d.**).

Interaction: Uses a conversational flow that mimics a real doctor's interview (**IntuitionLabs, 2025**).

Accessibility: Meets WCAG 2.1 AA standards for users with visual or motor impairments.



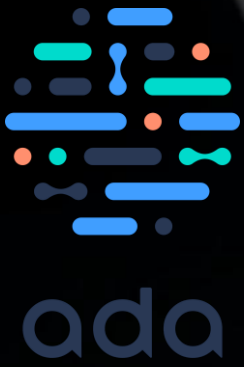
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SKILLS & THIRD PARTY INTEGRATIONS

Ada integrates with specific health insurers and health systems, thus expanding the capabilities and versatility of the system

Integration with wearables like smartwatches provides a holistic view of health data, such as heart rate or sleep patterns (**Droxy AI, 2025**)

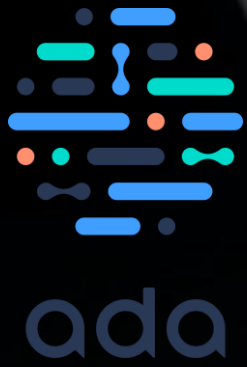
Integration with new devices (for example, MRI machines) expands the skills of the system



CONTINUOUS LEARNING

Ada uses a feedback loop where user interactions and physician feedback continuously refine its probabilistic reasoning engine

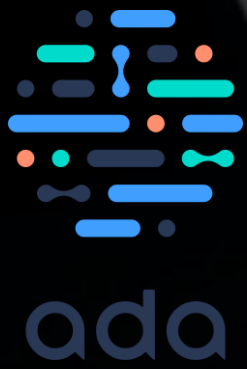
By analyzing historical preferences and contextual information, the system personalizes the conversational flow to provide accurate, real-time results (**Ada Health, n.d.; Shneiderman, 2022**).



INTERACTION TYPES

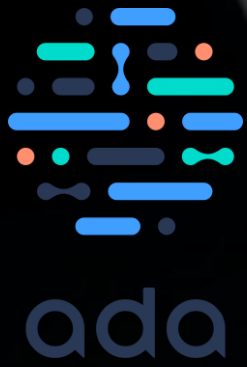
Familiar conversational chatbot interface with input over text or voice

By adhering to **WCAG 2.1 Level AA standards**, the system ensures accessibility for users with visual, auditory, or motor disabilities through screen readers and keyboard-only controls (**Ada.cx, n.d.**).



USER-CENTRIC DESIGN

In Human Centred Design, the user is a co-creator. Ada achieves this through plain language, empathetic conversational flow, and providing rationales behind advice to build trust (**Tzimourta et al., 2025**).

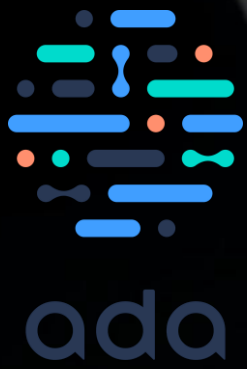


PRIVACY & SECURITY

Ada employs a "Privacy by Design" architecture

User identity is stored separately from health information and linked only by a unique key.

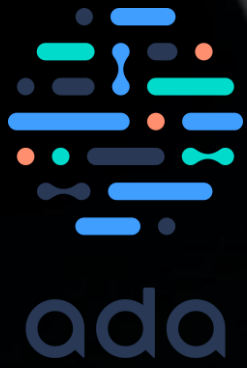
In the event of a breach, medical history cannot be easily tied to a real-world identity (**Ada Health, n.d.**).



SCALABILITY & SUSTAINABILITY

Scalability: To handle global demand, Ada utilizes managed cloud services to scale to global demand (AWS, 2025)

Sustainability: Ada's Global Health Initiative (GHI) provides free access in underserved regions like India and Africa, ensuring AI benefits are not limited to high-resource settings (AI4SDGs, n.d.).



CHALLENGES & FUTURE DIRECTIONS



Hi Julia. I can help you
figure out what's going
on. Let's start a
symptom assessment.

Start symptom assessment

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LESSONS LEARNED

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A critical lesson is the need for a "systems-first" approach.

Developers have found that chatbots fail if they lack integration with local cultural contexts or health systems **(ICTworks, 2026)**.

Automation Bias highlights the need for "Human-in-the-Loop" designs to prevent over-reliance on AI **(NIH, 2025)**.

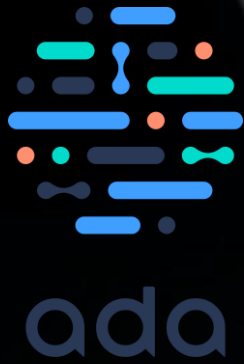


DEVELOPER CHALLENGES

The NLP must bridge the "Semantic Gap" between subjective patient descriptions and clinical terminology

Developers must navigate a fragmented global regulatory landscape (**Transparency International, n.d.**).

To maintain trust, they must solve the "Black Box" problem—ensuring deep learning models remain transparent and explainable (**Gupta, 2024; World Economic Forum, 2025**).



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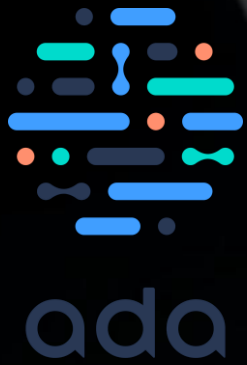
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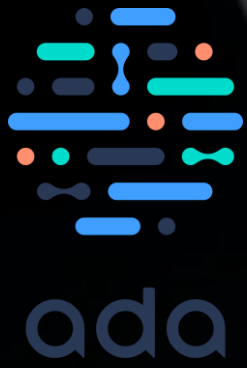
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AI Usage Statement: AI tools were used to assist with idea generation, drafting, and reference refactoring. Final decisions about content and structure were my own.

How did you use AI tools in this Assignment? I decided to try Gemini. First I told it about the assignment deliverables, and then I fed the discussion points into the chatbot one by one. I then rephrased the output into my own words.

How did AI influence or support your learning and decision making? It supported my learning by giving me new insights into what challenges a HCI system such as this has. I can now make a more informed decision when contemplating use of the chatbot for my own needs.



Hi Julia. I can help you
find out what's going
on. Just start a
symptom assessment.

Start symptom assessment

Questions?

ada